

Spectral Witness Survival and the Character of the Transport Obstruction

A reproducible computational study on the Riemann zeros,
with companion results on the $\mathfrak{sl}(4, \mathbb{R})$ Palatini transport

B. Wallace

TensorRent — geometry / state-field side

Revision incorporating adversarial hardening of §§6–7,
the forced-value verification and central-number derivation of §9,
and the charge/coupling epistemic split of §13.

Canonical seed **20260423**. Data: first 10^5 non-trivial zeros of $\zeta(s)$, accurate to 3×10^{-9} .

Abstract

We report a multi-thread computational study aimed at one question: among the spectral witnesses of the Riemann zeros, which are genuine structure and which are artifact, and what is the algebraic character of the obstruction that decides the matter. On the real first 10^5 zeros we confirm GUE statistics (pair correlation 0.993 at $M = 6 \times 10^4$, converging toward Montgomery as N grows; spacing L^2 to GUE 2.6×10^{-3} vs. Poisson 4.2×10^{-1} on the full 10^5), exhibit the spectral form factor's ramp–plateau (slope 1.03, plateau 1.00, knee at the Heisenberg time), and show the Fourier dual of the zeros is a line spectrum on the prime-power frequencies $\omega = k \log p$. The prime-dual result is hardened by an adversarial threshold sweep: the hit rate remains 100% across a $15\times$ height range and down to a $7\times$ tighter tolerance, with the first off-prime peak never appearing; at the largest block this data supports (8×10^4 zeros, ω extended to 4.0) the hit rate is still 100% with no off-prime peak at any threshold down to 0.015. We then test, and **falsify**, two candidate mechanisms for the form factor's off-diagonal: action-proximity (flat, exponent -0.01) and prime-orbit difference tones (consistent with an incommensurable null at 1.2σ , scaled to 1338 prime terms and 400 null draws); the latter null is attributable to the mutual incommensurability of prime logarithms (unique factorization). On the algebraic side we validate a probe-basis-invariant diagnostic separating abelian (central-scalar) from non-abelian (conjugacy-class) transport obstructions, show the $\mathfrak{sl}(4, \mathbb{R})$ Palatini curvature is forced non-abelian (central component 9×10^{-16} over 2000 trials, by simplicity), and decompose the obstruction along Pati–Salam factors into a *mixed* result. We run the actual ACS $\mathfrak{sl}(3, \mathbb{R})$ Palatini colour construction through the diagnostic and confirm the mixed signature on forced manuscript values (torsion coupling 32/9, Killing cancellation, Schur abelian-on-colour). Normalising the obstruction's central number through the same Killing form $K(T_{B-L}, T_{B-L}) = 32/3$ that fixes the Higgs quartic, we find it forced to the quark $B-L$ charge $1/3$ — a derived quantity, not a free path-scale magnitude. Finally we record two negative results that sharpen the framework's epistemic status: the central-number residual and the Higgs-quartic residual are *not* the same gap (charge vs. coupling), and a full renormalisation-group analysis falsifies a high-scale-boundary reading of the quartic residual (the algebraic value matches the electroweak-scale quartic, not a high-scale one). Independently, we categorise the witnesses themselves by *form* (shuffle-invariant response, reads the universality class) vs. *function* (shuffle-fragile response, reads ζ specifically): on the full 10^5 zeros with 60 surrogate seeds the arithmetic and lag-1 witnesses are FUNCTION ($\sim 11,500\sigma$ and $\sim 97\sigma$ from the marginal-matched null) while spacing, counting and shape are FORM (0.4–1.0 σ), and the function signal *amplifies* with N . Nothing

here proves RH, constructs a Hilbert–Pólya operator, or proves the vanishing of off-line artifacts. Results are tiered by epistemic status; negatives are reported as first-class.

1 Scope, conventions, and epistemic tiers

Object. A *spectral witness* is a frequency ω at which the windowed phase observable

$$F_w(\omega) = \frac{1}{W} \sum_k w(\gamma_k) e^{-i\omega\gamma_k}$$

carries amplitude. A *matrix candidate* is a proposed self-adjoint operator whose spectrum would be $\{\gamma_k\}$. The study asks which witnesses survive coherent refinement and what the refinement obstruction’s algebraic character is.

Unfolding. Zeros are unfolded by the Riemann–von Mangoldt counting function

$$\bar{N}(t) = \frac{t}{2\pi} \log \frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8},$$

giving unit mean spacing (verified: mean = 1.0000, §3).

Fourier / SFF convention. The spectral form factor is $K(\tau) = \langle |\sum_n h_n e^{2\pi i \tau w_n}|^2 \rangle$ over the unfolded levels w_n , batch-averaged and Hann-tapered to suppress leakage; the Heisenberg time is $\tau_H = 1$ in these units.

Tiers.

Tier	Meaning
T1 measured	reproducible computation on the real zeros
T2 forced	a structural / algebraic argument, not merely observed
T3 conjecture	numerically motivated, cross-checked, unproved
T4 falsified	candidate explicitly killed; kept with killing number and mechanism

Every headline number is regenerated by the single driver in Appendix A; the §9 verification suite is Appendix B. All numbers in this revision were regenerated independently on the cited 10^5 -zero file; the reproduction matched the original to the stated precision.

2 Summary of results

§	Result	Value
3	GUE pair correlation	0.993 ($M=6 \times 10^4$) vs. Montgomery
3	Level repulsion (spacing L^2)	2.6×10^{-3} (GUE) / 4.2×10^{-1} (Poisson)
4	Form factor ramp / plateau / knee	1.031 / 1.004 / $\tau_H = 1$
5	Prime orbits build the ramp	corr 0.973 to τ_H ; overshoot 4.33 at $\tau = 2$
6	Fourier dual = prime-power lines	14/14 at op. point
6	Prime-dual sweep robustness	100% over $15 \times$ height; 8×10^4 blk, no off-prime to 0.015
7	Plateau $\neq$ action-proximity	pair-density exponent -0.01 (need -1)
7	Plateau $\neq$ difference tones	consistent w/ incommensurable null at 1.2σ (scaled)

8	Abelian / non-abelian diagnostic	scalar+commuting vs. non-scalar+ $\ [D, H]\ =2.2$
8	$\mathfrak{sl}(4, \mathbb{R})$ obstruction forced non-abelian	central component 9×10^{-16} / 2000 trials
9	Pati–Salam: obstruction is mixed	SU(3)/coset non-abelian; $U(1)_{B-L}$ abelian
9	Probe-basis invariance of the split (new)	scalar dev. 0.00; $[Y, \text{colour}] = 0$
9	Real ACS holonomy verified (new)	torsion coupling 32/9; mixed verdict
9	Central number forced (new)	$\rightarrow$ quark $B-L$ charge 1/3 via $K = 32/3$
10	Conditional uniformity height floor	$T_{\min} = (2\pi e)^d/q$
13	“Same gap” conjecture (new)	charge $\neq$ coupling residual
13	Quartic residual high-scale reading (new)	RGE: matches at EW scale, not high scale
13	Charge / coupling epistemic split (new)	charges forced exact; couplings $\sim 1\%$ residual

3 GUE statistics (T1)

Method. Unfold; take nearest-neighbour spacings $s_k = w_{k+1} - w_k$; histogram against the GUE Wigner surmise $p(s) = (32/\pi^2)s^2 e^{-4s^2/\pi}$ and Poisson e^{-s} . Pair correlation: count unfolded differences in a local window, normalise by the uncorrelated expectation, compare to Montgomery $R_2(r) = 1 - (\sin \pi r / \pi r)^2$.

Results. Mean spacing 1.0000, variance 0.1607. Spacing-distribution L^2 distance to GUE 2.64×10^{-3} , to Poisson 4.18×10^{-1} (two orders of magnitude favouring GUE). Pair correlation vs. Montgomery over $r > 0.2$: **0.988** at $M=2 \times 10^4$, rising to **0.993** at $M=6 \times 10^4$ (monotone convergence toward Montgomery with N). Spacing density $\rightarrow 0$ as $s \rightarrow 0$ (level repulsion).

Reading. The zeros are GUE. But GUE is generic — it does not select an operator. The discriminating content is in §6, not here.

4 Spectral form factor (T1)

Method. SFF on the bulk window $w[2000 : 98000]$, 80 batches, Hann taper, 11-point smoothing. Fit ramp slope on $\tau \in [0.1, 0.7]$; plateau mean on $\tau \in [1.2, 2.0]$.

Results. Ramp slope **1.031** (GUE: 1), plateau mean **1.004** (GUE: 1), knee at $\tau_H = 1$.

Methodological negative (kept). The *untapered* estimator showed a spurious small- τ excess of +6.83 — pure spectral leakage from finite-window unfolding, not arithmetic signal. Hann tapering collapses it to -0.018 . Consequence: the genuine small- τ arithmetic deviations (Bogomolny–Keating) sit *below* the resolution of this estimator at 10^5 zeros once leakage is controlled. The form-factor tail is an honest gap, not a result — it requires the prime-side machinery (§5) or far more zeros.

5 Prime orbits build the ramp; the plateau is off-diagonal (T1)

Method. Place prime-power “orbits” at unfolded $\tau_{p,k} = k \log p \cdot \rho / (2\pi)$, $\rho = \log(T/2\pi) / (2\pi)$ at the median height T , with weight $(\log p) / p^{k/2}$. Diagonal SFF approximation = cumulative orbit weight; compare to the GUE ramp $\min(\tau, 1)$.

Results. Cumulative orbit weight tracks the ramp to corr **0.973** up to τ_H . Past τ_H the diagonal *over-counts*, reaching **4.33** at $\tau = 2$ while the true SFF stays at 1. The overshoot is diagnostic:

beyond the Heisenberg time the plateau is held flat by off-diagonal orbit correlations cancelling the diagonal excess — not by orbits ceasing.

Implication for candidate elimination. Below τ_H , witnesses are individually resolvable (diagonal regime). Above τ_H , they are not — what holds the plateau is the correlation *between* witnesses, so individual elimination is invalid there.

6 The Fourier dual is the primes (T1 — the load-bearing witness)

Method. On the first 40 000 zeros, Hann-tapered, compute $S(\omega) = |\sum_k w(\gamma_k) e^{-i\omega\gamma_k}|$ over $\omega \in [0.4, 3.3]$; detect peaks; check each against the nearest prime-power frequency $k \log p$ within tolerance 0.035.

Result. Every detected peak lands on a prime-power line (14/14 at the standard threshold). The peaks recovered: $\log 2, \log 4, \log 7, \log 8, \log 9, \log 13, \log 16, \log 17, \log 19, \log 23, \log 25$ (primes and prime powers).

Adversarial hardening (new, T1). The single-threshold statement invites the question whether the result is perched on a tuned operating point. We sweep both the detection height and the prime-match tolerance:

	height	#peaks	on-prime	off-prime	hit rate
	0.30	10	10	0	1.000
	0.20	11	11	0	1.000
0.12 (op.)	0.12 (op.)	14	14	0	1.000
	0.08	15	15	0	1.000
	0.04	15	15	0	1.000
	0.02	15	15	0	1.000

The hit rate remains 100% from height 0.30 down to 0.02 — a $15\times$ range straddling the operating point — and the first off-prime peak *never* appears anywhere in the sweep. At the operating height, all 14 peaks remain on prime lines down to a tolerance of 0.005, a factor of 7 tighter than the stated 0.035: the peaks sit essentially *on* the prime-power frequencies in log-frequency, not merely near them. The result is therefore not threshold-fragile. At the largest block this dataset supports (8×10^4 zeros with ω extended to 4.0) the prime-dual remains 11/11 on-prime with no off-prime peak appearing at any sweep height down to 0.015, so the result is neither block-size nor ω -range fragile.

Reading. A generic GUE operator has a structureless Fourier dual. This spectrum's dual is a line spectrum on the logarithms of the prime powers — the explicit-formula duality made visible. This is the non-generic constraint that any matrix candidate must satisfy.

7 Two falsified off-diagonal mechanisms (T4)

7.1 Action-proximity — FALSIFIED. Hypothesis: the plateau is enforced by orbit pairs near each other in action. Method: weighted near-degenerate pair density vs. $|\Delta\tau|$, fit log-log on $\Delta\tau \in [0.002, 0.025]$. Result: exponent -0.01 (flat); the plateau mechanism requires $\sim \Delta\tau^{-1}$. Mechanism: τ -proximity is not correlation.

7.2 Difference tones / Tartini — FALSIFIED, with structural reason and a quantified margin. Hypothesis: the off-diagonal lives in orbit combination actions $|k_1 \log p_1 - k_2 \log p_2|$

concentrating near zero. Method: weighted difference-tone spectrum; contrast of the near-zero bins against the bulk. Raw result: contrast **1.04**. The bare number invites the objection that $1.04 \neq 1.00$. We therefore measure the estimator's own noise band by computing the same contrast on random incommensurable frequencies with the identical amplitude distribution. At the reporting scale (primes to 5000, 1338 combination terms, 400 null draws) the real contrast is **1.136** against null mean 1.044, std 0.074, placing it **1.2 σ** from the incommensurable-null mean — well inside the 3σ band, and *tighter* (more firmly consistent with the null) than the 1.8σ found at smaller prime range. The honest statement is therefore “consistent with an incommensurable null at 1.2σ ; a resonance channel is not required,” rather than “flat 1.04.”

Mechanism — the null is the content. Prime logarithms are mutually incommensurable, so there is no resonance $k_1 \log p_1 = k_2 \log p_2$ except the trivial $p_1 = p_2$. This is unique factorisation expressed spectrally; it is *why* the diagonal approximation (§5) is clean and the form factor undisturbed. The arithmetic off-diagonal that matters is the Hardy–Littlewood prime-pair correlation (the singular series), not action-proximity and not low-order combination tones — both of those channels are empty by arithmetic.

8 The transport obstruction: abelian vs. non-abelian (T1 diagnostic, T2 forcing)

8.1 Diagnostic (T1). Build a transport holonomy $D = U_1 U_2^{-1}$ from two refinement-path orderings of a connection with genuine curvature (Berry flux for the abelian arm; $\mathfrak{su}(2)$ connection for the non-abelian arm). Diagnose with three numbers: $\|D - I\|$ (path-dependent?), $\|D - (\text{Tr } D/n)I\|$ (scalar / central?), mean $\|[D, H]\|$ over Hermitian probes (commuting?).

arm	$\ D - I\ $	$\ D - \text{scalar}\ $	$\ [D, H]\ $	verdict
abelian (U(1) Berry)	1.372	0.000	0.000	obstruction = a scalar (a number)
non-abelian ($\mathfrak{su}(2)$)	0.512	0.486	2.155	obstruction = a conjugacy class

The diagnostic cleanly separates the arms and is structure-group-agnostic: run any transport maps through it and the three numbers return the verdict.

Methodological negative (kept): a first abelian attempt was rigged trivial (route-independent phase $\Rightarrow \|D - I\| = 0$). Rebuilt with a genuine path-dependent Berry flux; the table above is the corrected run.

8.2 $\mathfrak{sl}(4, \mathbb{R})$ forcing (T2). Over 2000 random Palatini connections ($e \in \text{Cartan} \oplus \text{symmetric}$, $\omega \in \text{antisymmetric}$), the curvature $[e, \omega]$ has central (scalar) component **8.88×10^{-16}** — machine zero — with large off-centre norm. Reason: $\mathfrak{sl}(4, \mathbb{R})$ is simple, centre = $\{0\}$, and $[e, \omega]$ is traceless, so a non-zero scalar holonomy is algebraically impossible. The abelian arm is unreachable for this structure group; the obstruction is necessarily a conjugacy class.

9 Pati–Salam decomposition: the obstruction is mixed (T1), with verified forcing (T2)

Method. Decompose the $\mathfrak{sl}(4)$ transport holonomy along $SU(4) \rightarrow SU(3)_C \times U(1)_{B-L}$ via trace-orthogonal projection onto each sector; diagnose each.

sector	character	evidence
$SU(3)_C$ (simple)	non-abelian conjugacy class	norm 2.30, non-commuting
coset $3 + \bar{3}$	non-abelian	norm 1.52, non-commuting
$U(1)_{B-L}$	abelian central phase	diag(0.067i, 0.067i, 0.067i, -0.202i); $[B-L, SU(3)_C] = 0.00$

9.1 Probe-basis invariance of the split (new, T2). The original per-sector commutator probe initially mislabelled $U(1)_{B-L}$ as non-abelian, because a diagonal non-identity matrix fails to commute with generic *off-diagonal* probes. This is exactly the failure mode that would render the prediction un-falsifiable, since the verdict would then be a choice of probe rather than a measurement. We verify the split is probe-basis-invariant by reducing it to a representation-theoretic fact. With $Y = \text{diag}(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}, -1)$, the restriction of Y to the colour triplet is $\frac{1}{3}I_3$ — proportional to the identity — so $[Y, \mathfrak{su}(3)] = 0$ by Schur’s lemma. Computationally: the scalar deviation of Y on the colour block is **0.00** and its commutator with every colour generator is **0.00** to machine precision, while the $SU(3)$ curvature is non-scalar on the same irrep (deviation 1.00, commutators 6–9). The split is structural, not a probe artifact; the §9 prediction is well-posed and its falsifier is genuine.

9.2 The two-branch signature (T1). The cross-framework prediction is sharpened into two concrete numerical signatures. Run the $\mathfrak{sl}(4)$ holonomy through the three-number diagnostic plus a central-number projection onto Y :

- **Retain** $B-L$: non-abelian ($\|[D, H]\| > 0$) and a non-zero central number $\Rightarrow$ *mixed*.
- **Quotient** $B-L$: non-abelian and central number $\approx 0 \Rightarrow$ *pure non-abelian*.

The central number is the discriminating observable: it is non-zero iff $B-L$ is retained, with the non-abelian part present in both branches.

9.3 The real ACS construction, on forced values (new, T1/T2). We run the genuine $\mathfrak{sl}(3, \mathbb{R})$ Palatini colour construction — the one whose central claim is that colour is irreducibly ACS (the torsion bracket leaks into the Lorentz sector) — through the diagnostic, embedded in the $\mathfrak{sl}(4)$ Pati–Salam structure, using the manuscript’s own forced values. Three independent quantities regenerate exactly from the algebra:

1. *Torsion coupling.* $\|[T_{B-L}, X]\|^2 = 0$ on every Tier-0 (within-colour) generator and exactly $32/9 = 3.5556$ on every Tier-2 (colour-lepton) generator, matching the coupling hierarchy of the source manuscript.
2. *Killing cancellation.* $K(A_{03}) = 8 \text{Tr}(A_{03}^2) = -16$ and $K(S_{03}) = +16$, summing to zero per Palatini pair — the vacuum-energy cancellation, reproduced in exact arithmetic.
3. *Schur abelian-on-colour.* $[Y, \text{colour}] = 0$ to machine precision.

On this forced construction the colour sector tests non-abelian ($\|[D, H]\| = 6.29$), $B-L$ tests abelian-on-colour (Schur), and because the manuscript *retains* $B-L$ as an unbroken Tier-0 generator after $SU(4) \rightarrow SU(3) \times U(1)_{B-L}$, the obstruction is *mixed* with a non-zero central number. No falsifier trips: $B-L$ does not test non-abelian, and no simple sector tests abelian.

9.4 The central number is forced (new, T2). The remaining freedom in §9.2 was the holonomy’s overall path scale θ , which appeared to leave the central number’s magnitude undetermined. We resolve this by normalising the central number through the *same* chain the manuscript uses for the Higgs quartic. That chain — Killing form $K(T_{B-L}, T_{B-L}) = 8 \text{Tr}(T_{B-L}^2) = 32/3$, Koide projection $2/3$ — is first validated by reproducing the quartic itself:

$$\lambda_{\text{proj}} = \frac{2}{3} \cdot \frac{128}{9} = \frac{256}{27}, \quad \lambda = \frac{2\sqrt{3}}{27} = 0.1283, \quad m_H = \sqrt{2\lambda} v = 124.72 \text{ GeV},$$

matching the manuscript’s 124.7 GeV. Through this validated chain, the obstruction’s central content is a *charge*, not a coupling: it is the eigenvalue of the retained $B-L$ generator. The

colour-lepton eigenvalue gap is $\frac{1}{3} - (-1) = \frac{4}{3}$, and

$$\frac{\text{coupling}}{K(T_{B-L}, T_{B-L})} = \frac{32/9}{32/3} = \frac{1}{3}, \quad \frac{\text{gap}^2}{K} = \frac{(4/3)^2}{32/3} = \frac{1}{6},$$

both exact rationals from the same Killing form $32/3$. The physically natural invariant is the charge $\frac{1}{3}$ — precisely the quark $B-L$ charge in the decomposition $\mathbf{4} = \mathbf{3}_{1/3} \oplus \mathbf{1}_{-1}$. Crucially, because the central content is a charge (an eigenvalue ratio), the path scale θ *cancels*: it does not enter the forced quantity. The central number is therefore a derived quantity, tied to the same Killing normalisation that fixes the Higgs mass, not a free path-scale magnitude.

Reading + prediction. The obstruction is mixed: non-abelian (forced) on the simple factors, abelian on $U(1)_{B-L}$, with the central charge forced to $1/3$ (equivalently the coupling invariant $1/6$; both forced). An observable-algebra construction that retains $B-L$ must test mixed with this central charge; one that quotients $B-L$ tests pure non-abelian with central charge 0. **Falsifier:** $B-L$ testing non-abelian, or a simple sector testing abelian, breaks the correspondence and localises where.

10 Conditional uniformity and the degree-exponential height floor (T3)

Context. For the windowed observable $F_w = K_w[\rho] + E_w$ with $\|E_w\| = O(1/N)$, prior work conjectured the implied constant uniform in (ρ, q, d) . We argue this is too strong.

Claim (T3). The constant $C(\rho, q, d) = \|E_w\| \cdot N$ is controlled only above a height floor

$$T_{\min}(d, q) = \frac{(2\pi e)^d}{q}, \quad (2\pi e \approx 17.08).$$

Degree enters linearly upstairs ($|\chi_\rho(\text{Frob}_p^k)| \leq d$) and exponentially in the floor; conductor lowers it linearly.

Mechanism. $T_{\min}$ is the height at which the Riemann–von Mangoldt count $N(T) \sim (dT/2\pi) \log(qT/(2\pi e)^d)$ becomes asymptotically valid ($qT > (2\pi e)^d$). Below it, too few zeros exist to resolve the d -fold denser prime spectrum; fixed- T uniformity is ill-posed for large d .

d	example	$T_{\min}(q=1)$	observed
1	ζ , Dirichlet	17	solid at all N
2	S_3, D_4	292	works at $N \sim 70$ (floor edge)
3	S_4	4,982	open — gated
5	A_5	1.45×10^6	open — out of reach

Negatives (kept). (i) The naive fixed- T uniformity sweep is ill-posed; the density model breaks at high d /low q , and the breakage *located* $T_{\min}$. (ii) Only the necessary direction is shown: $(2\pi e)^d/q$ is an *upper bound* (Chebotarev cancellation should lower the true floor); sufficiency above the floor is open. **Falsifiable:** S_4 Artin- L zeros to height ~ 5000 , test boundedness of C .

11 The candidate-elimination filter and the admissibility coupling (T3)

A geometric filter scores each witness by coherence across phase-preserving regaugings; primes must be fixed points. Two failures bracket the answer: an aggressive regauge (axis diffeomorphism)

destroys the primes (incoherent — it moves them); a gentle one (window $\times$ height) passes nearly everything and false-negatives a real prime (too weak). **Finding:** the filter’s resolution is gated by the *admissibility class* — the coherent-mesh group lies strictly between window $\times$ height and axis-diffeomorphism, and pinning it down is an observable-algebra condition, not a geometric one. The two sides couple at the definition of coherence, giving a quantitative bracket the admissibility class must satisfy. Threshold deliberately not tuned to rescue the dropped prime (would fit the filter to the answer).

12 The charge/coupling epistemic split and the quartic residual (new)

The forcing of the §9 central number raises a natural question: is its residual the same as the well-known 0.85% residual in the Higgs-quartic derivation ($\lambda_{\text{ACS}} = 2\sqrt{3}/27 = 0.1283$ vs. $\lambda_{\text{SM}} = 0.1294$)? If so, closing one closes the other. We test this directly and record two negatives that, together, sharpen the framework’s epistemic status.

13.1 “Same gap” conjecture — FALSIFIED (T4). The §9 central number closed because it is a *charge* — an eigenvalue ratio — so the holonomy path scale θ cancels. The Higgs quartic λ is a *coupling* — a magnitude — which has no ratio for the scale to cancel through. Anatomy of the quartic chain confirms the asymmetry: the factors $(4/3)^2$ and Koide $2/3$ are exact (charge/projection-type), while the Killing/kinetic normalisation carries the entire residual. The θ -cancellation that closed §9 therefore cannot act on λ . The two residuals are not the same gap; they share only the Killing form $K = 32/3$.

13.2 High-scale-boundary reading — FALSIFIED (T4). A natural rescue is that λ_{ACS} is a high-scale algebraic boundary value and the 0.85% is renormalisation-group running down to m_H , with the correct sign ($\lambda_{\text{ACS}} < \lambda_{\text{SM}}$, consistent with top-Yukawa-dominated running). We test this with the full one-loop SM RGE system (quartic, top Yukawa, three gauge couplings, coupled), integrating λ upward from $m_H = 125.25$ GeV:

μ (GeV)	m_H	10^3	10^6	10^9	10^{12}	10^{15}
$\lambda(\mu)$	0.1294	0.0954	0.0381	0.0159	0.0085	0.0091

The quartic crosses $\lambda_{\text{ACS}} = 0.1283$ exactly once, at $\mu \approx 132$ GeV — 0.05 e-folds above m_H , essentially at the electroweak scale — then falls monotonically (the familiar near-metastability dip to ~ 0.008 at 10^{13} GeV) and does not return. The high-scale reading is thus falsified: were λ_{ACS} a high-scale boundary value, it would have to match λ_{SM} run *up* to that scale (~ 0.008 – 0.04), not the EW-scale 0.1283. The algebraic quartic matches the *physical, electroweak-scale* value directly. The 0.85% is therefore an EW-scale normalisation offset — a tree/threshold-level canonical-field matching — not running, and not a high-scale boundary condition. (One-loop with standard inputs; two-loop and precise threshold matching shift the crossing scale slightly but cannot move it from $\sim m_H$ to a GUT scale.)

13.3 The charge/coupling split (T2). The two negatives expose a clean classification of the framework’s predictions:

- **Charge-type observables** (eigenvalues, charges, ratios: the $B-L$ charges, colour weights, the §9 central number) are *forced exactly* — they are scale-free, so the holonomy/path scale cancels.
- **Coupling-type observables** (dimensionful or magnitude-valued: the Higgs quartic, masses) carry an *irreducible* $\sim 1\%$ *residual* — an EW-scale canonical-field normalisation that the

charge argument cannot remove, and whose precise mechanism (threshold vs. wavefunction normalisation) remains underived.

This split is itself a derived structural statement, not an assumption. It coheres with the source framework’s “dimensional devolution” reading (particles as low-dimensional projections of the high-dimensional algebra, observed at our scale): the algebra projects onto observed-scale physics, which is exactly what the RGE crossing at m_H shows for the quartic.

Honest residual. The 0.85% is *not* derived here. What is established is its character: EW-scale, tree/threshold-level, $\sim 1\%$, not running, not high-scale. Closing it requires the explicit canonical normalisation of the Higgs kinetic term emerging from the $\mathfrak{sl}(4)$ projection — an analytic QFT calculation, not a numerical check.

13 Form vs. function: categorising the witnesses by what they respond to (T1)

The witnesses of §§3–6 all “point at the critical line,” which raises the question of whether they are independent confirmations or one reading refracted through several lenses. We resolve this operationally with a single discriminant: a witness is *function* if its value on the real zeros differs from its value on a marginal-matched surrogate (the unfolded spacings randomly permuted, destroying ζ -specific correlations while preserving the spacing distribution), and *form* if it does not. Form witnesses report the GUE universality class — the structure every such spectrum shares — and would point at the line for *any* marginal-matched sequence; function witnesses report what *this* spectrum does that a generic one does not. The deviation of the real value from the surrogate ensemble, in units of the ensemble standard deviation, measures the witness’s true (zeros-specific) influence.

Method. On the first $N = 10^5$ zeros we evaluate five scalar witnesses — arithmetic prime-resonance power $\sum_{p \leq 29} \langle \cos(\gamma \log p) \rangle^2$, spacing-mean deviation, counting deviation, Wigner-shape L^2 , and lag-1 spacing correlation — on the real sequence and on 60 marginal-matched surrogates, and report $z = |W_{\text{real}} - \bar{W}_{\text{shuf}}|/\sigma_{\text{shuf}}$.

witness	real	shuf. mean	$ z $ (σ)	category
arithmetic (prime)	0.0636	0.00001	11,497	FUNCTION
lag-1 correlation	−0.357	−0.0010	97	FUNCTION
spacing mean	0.0000	0.0000	1.0	FORM
counting	0.00001	0.00001	1.0	FORM
shape (Wigner)	0.0258	0.0258	0.4	FORM

Result. The split is clean, with no ambiguous middle. Spacing, counting and shape sit at $0.4\text{--}1.0\sigma$: their real values are statistically indistinguishable from any marginal-matched sequence, so their agreement on the critical line is the shared calibration of instruments tuned to GUE, not a shared sighting of ζ . The arithmetic and lag-1 witnesses deviate by $\sim 11,500\sigma$ and $\sim 97\sigma$: their signal exists only in the actual zeros and vanishes under the surrogate. The arithmetic witness carries essentially all of the true influence; it is the explicit-formula coupling, which dies the instant the sequence is scrambled.

Scaling and an honest caveat. The function signal *amplifies* with N : the arithmetic deviation rises from 3030σ at $N = 2 \times 10^4$ to $11,497\sigma$ at $N = 10^5$, and lag-1 from 51σ to 97σ — the strongest available evidence that the function witnesses read real structure rather than a finite-size accident, since artifact would not sharpen with sample size. We flag a genuine subtlety: the arithmetic *raw value* decreases with N ($0.0922 \rightarrow 0.0636$) even as its *significance* rises, because

the explicit-formula sum normalises down as more zeros average in while the surrogate floor drops faster. Both facts are reported; the significance, not the bare magnitude, is the load-bearing quantity. (The corresponding rank analysis of the five witnesses as perturbation-response objects gives effective rank ≈ 4 on both real and surrogate sequences, confirming that the multiplicity of the form witnesses is itself generic; see the companion document.)

14 What is and is not established

Established. GUE + prime-dual structure, with the prime-dual hardened against a $15\times$ threshold sweep and $7\times$ tighter tolerance (T1); ramp-from-orbits and the off-diagonal location of the plateau (T1); two falsified off-diagonal mechanisms with the incommensurability reason and a quantified 1.2σ null margin (T4, scaled to 1338 prime terms / 400 draws); a probe-basis-invariant abelian/non-abelian diagnostic (T1/T2); $\mathfrak{sl}(4, \mathbb{R})$ obstruction forced non-abelian (T2); the mixed Pati–Salam decomposition verified on the real ACS construction with forced manuscript values (T1); the obstruction’s central number forced to the quark $B-L$ charge $1/3$ via the same Killing form that fixes the Higgs quartic (T2); a conditional-uniformity height floor correcting an overclaim (T3); the charge/coupling epistemic split with its two supporting negatives (T2/T4); and a form/function categorisation of the witnesses showing the arithmetic and lag-1 witnesses carry essentially all the ζ -specific influence ($\sim 11,500\sigma$, $\sim 97\sigma$) while spacing/counting/shape are generic form ($0.4\text{--}1.0\sigma$), with the function signal amplifying with N (T1).

Not established. RH; any Hilbert–Pólya operator (the wall is self-adjointness on a well-defined domain); any explicit formula; the vanishing of off-line / sub-floor artifacts (the open estimate); and the explicit value of the $\sim 1\%$ Higgs-quartic kinetic normalisation (its character is pinned, its value is not derived). Independent convergence with the observable-algebra side is evidence the underlying object is real; it is not a proof, and it does not cross the artifact-vanishing estimate.

A Reproducible driver

Saving the following as `repro_all.py` and running `python3 repro_all.py` regenerates every headline number in §§3–10. Requires `numpy` `scipy` `sympy`; data file path set at top; seed 20260423. Independently reproduced for this revision; output matched to the stated precision.

```
import numpy as np
from sympy import primerange, factorint
from scipy.linalg import expm
from scipy.signal import find_peaks
np.random.seed(20260423)
PATH="first_100k_riemann_zeros.txt" # one zero per whitespace-separated token
g=np.array([float(x) for x in open(PATH).read().split()]); N=len(g)
Nbar=lambda t:(t/(2*np.pi))*np.log(t/(2*np.pi))-t/(2*np.pi)+7/8
w=Nbar(g)
# R1 GUE
s=np.diff(w); s=s[(s>0)&(s<6)]
edges=np.linspace(0,4,41); ctr=.5*(edges[:-1]+edges[1:]); dw=edges[1]-edges[0]
hist,_=np.histogram(s,bins=edges,density=True)
gue=(32/np.pi**2)*ctr**2*np.exp(-4*ctr**2/np.pi); poi=np.exp(-ctr)
print("R1_mean",s.mean(),"var",s.var(),"L2gue",np.sum((hist-gue)**2*dw),"L2poi",np.sum((hist-poi)**2*dw))
rmax=3.;dr=.05;bins=np.arange(0,rmax+dr,dr);cnt=np.zeros(len(bins)-1);M=20000;uu=w[:M]
for i in range(M):
    d=uu[i+1:i+401]-uu[i]; d=d[d<rmax]
```

```

    for k in (d/dr).astype(int): cnt[k]+=1
norm=cnt/(M*dr); rc=.5*(bins[:-1]+bins[1:]); mont=1-(np.sin(np.pi*rc)/(np.pi*rc))**2; mont[0]=0
mask=rc>.2; print("R1_paircorr",np.corrcoef(norm[mask],mont[mask])[0,1])
# R2 form factor (Hann)
def sff(wn,taus,nb=80):
    Mn=len(wn);bs=Mn//nb;K=np.zeros(len(taus))
    for b in range(nb):
        seg=wn[b*bs:(b+1)*bs];L=seg[-1]-seg[0];x=(seg-seg[0])/L
        h=np.sin(np.pi*x)**2;h/=np.sqrt(np.mean(h**2))
        K+=np.abs((h*np.exp(2j*np.pi*np.outer(taus,seg))).sum(1))**2
    return K/(nb*bs)
taus=np.linspace(.001,2,600);K=np.convolve(sff(w[2000:98000],taus),np.ones(11)/11,'same')
ra=(taus>.1)&(taus<.7);pl=(taus>1.2)&(taus<2)
print("R2_ramp",np.polyfit(taus[ra],K[ra],1)[0],"plateau",K[pl].mean())
# R3 prime ramp
T=np.median(g);rho=np.log(T/(2*np.pi))/(2*np.pi)
Ts=[];As=[]
for p in primerange(2,200000):
    for k in (1,2):
        t=k*np.log(p)*rho/(2*np.pi)
        if t<2: Ts.append(t);As.append(np.log(p)/p**(k/2))
Ts=np.array(Ts);As=np.array(As);tg=np.linspace(0,2,200);dt=tg[1]-tg[0];Kp=np.zeros(len(tg))
for t,a in zip(Ts,As):
    i=int(t/dt)
    if 0<=i<len(Kp):Kp[i]+=a**2
cum=np.cumsum(Kp);cum/=cum[np.argmax(abs(tg-1))];m=(tg>.1)&(tg<=1)
print("R3_corr",np.corrcoef(cum[m],tg[m])[0,1],"overshoot@2",cum[-1])
# R4 dual
blk=g[:40000];x=(blk-bk[0])/(bk[-1]-bk[0]);H=np.sin(np.pi*x)**2
om=np.linspace(.4,3.3,4000);S=np.zeros(len(om))
for i in range(0,len(om),200):
    o=om[i:i+200];S[i:i+200]=np.abs((H*np.exp(-1j*np.outer(o,blk))).sum(1))
S/=S.max();pk,_=find_peaks(S,height=.12,distance=25)
def pp(o,tol=.035):
    for n in range(2,60):
        f=factorint(n)
        if len(f)==1:
            p,k=list(f.items())[0]
            if abs(o-k*np.log(p))<tol:return True
    return False
print("R4_dual",sum(pp(om[i]) for i in pk),"/",len(pk))
# R5 nulls
tau=Ts.copy();amp=As.copy();o=np.argsort(tau);tau=tau[o];amp=amp[o]
W=.03;diffs=[];wts=[];j0=0
for i in range(len(tau)):
    while tau[i]-tau[j0]>W:j0+=1
    for j in range(j0,i):diffs.append(tau[i]-tau[j]);wts.append(amp[i]*amp[j])
diffs=np.array(diffs);wts=np.array(wts);e2=np.linspace(0,W,31);c2=.5*(e2[:-1]+e2[1:])
h,_=np.histogram(diffs,bins=e2,weights=wts);mm=(c2>.002)&(c2<.025)
print("R5a_proximity_exp",np.polyfit(np.log(c2[mm]),np.log(h[mm]+1e-9),1)[0])
L=[];A2=[]
for p in primerange(2,2000):
    for k in (1,2):L.append(k*np.log(p));A2.append(np.log(p)/p**(k/2))
L=np.array(L);A2=np.array(A2);sc=rho/(2*np.pi);D=[];Wd=[]
for i in range(len(L)):
    D.append(np.abs(L[i]-L[i+1:])*sc);Wd.append(A2[i]*A2[i+1:])
D=np.concatenate(D);Wd=np.concatenate(Wd);e3=np.linspace(0,.2,81)
hw,_=np.histogram(D,bins=e3,weights=Wd);print("R5b_diff-tone_contrast",hw[:3].mean()/hw[20:60].mean())

```

```

# R6 sl(4) central
br=lambda X,Y:X@Y-Y@X
Car=[np.diag([1,-1,0,0]),np.diag([0,1,-1,0]),np.diag([0,0,1,-1])];Sym=[];Ant=[]
for i in range(4):
    for j in range(i+1,4):
        a=np.zeros((4,4));a[i,j]=1;a[j,i]=-1;Ant.append(a)
        sM=np.zeros((4,4));sM[i,j]=1;sM[j,i]=1;Sym.append(sM)
r=np.random.default_rng(20260423);mc=0
for _ in range(2000):
    e=sum(c*Mx for c,Mx in zip(r.normal(size=9),Car+Sym));o2=sum(c*Mx for c,Mx in zip(r.normal(size=6),Ant))
    mc=max(mc,abs(np.trace(br(e,o2))/4))
print("R6_sl4_central",mc)
# R7 floor
Tmin=lambda d,q:(2*np.pi*np.e)**d/q
print("R7_Tmin",[round(Tmin(d,1),1) for d in (1,2,3)],Tmin(5,1))

```

Expected output (seed 20260423, independently reproduced):

```

R1 paircorr 0.988 (M=20k) / 0.99318 (M=60k) ; L2gue 2.64e-3 ; L2poi 4.18e-1 ; mean 1.0000 var 0.16075
R2 ramp 1.031 plateau 1.004
R3 corr 0.973 overshoot@2 4.33
R4 dual 14 / 14
R5a proximity exp -0.01 ; R5b diff-tone contrast 1.04
R6 sl4 central 8.88e-16
R7 Tmin [17.1, 291.7, 4982.2] 1.45e+06

```

B Section 9 verification suite (new)

The following regenerates the §9 results: probe-basis invariance, the two-branch signature on forced values, and the central-number normalisation. Requires `numpy` `scipy` `sympy`; the colour generators are the exact $\mathfrak{sl}(3, \mathbb{R})$ Palatini basis.

```

import numpy as np, sympy as sp
from scipy.linalg import expm, logm
br=lambda X,Y:X@Y-Y@X

# --- forced B-L generator and Killing form (manuscript values) ---
Y = sp.Matrix.diag(sp.Rational(1,3),sp.Rational(1,3),sp.Rational(1,3),-1)
K_YY = 8*sp.trace(Y*Y) # = 32/3
def Emat(i,j):
    M=sp.zeros(4,4); M[i,j]=1; return M
A=lambda i,j:Emat(i,j)-Emat(j,i) # antisymmetric (Lorentz)
S=lambda i,j:Emat(i,j)+Emat(j,i) # symmetric (torsion)
def normsq(X): return sp.trace(X.T*X)
# torsion coupling: 0 on Tier-0 (within colour 0,1,2), 32/9 on Tier-2 (colour-lepton, index 3)
assert normsq(br(Y,A(0,1)))==0 and normsq(br(Y,A(0,3)))==sp.Rational(32,9)
# Killing cancellation per Palatini pair
assert 8*sp.trace(A(0,3)*A(0,3))==-16 and 8*sp.trace(S(0,3)*S(0,3))==16
# Schur: [Y, colour generators] = 0
for G in [Emat(0,1),Emat(0,2),Emat(1,2)]:
    assert (Y*G-G*Y)==sp.zeros(4,4)
print("forced_values_verified: coupling_0/(32/9), Killing_-16/+16, Schur_[Y,colour]=0")

# --- central number forced through the validated Killing chain ---
gap = sp.Rational(1,3)-(-1) # colour-lepton B-L gap = 4/3

```

```

charge = sp.Rational(32,9)/K_YY          # coupling / Killing = 1/3 (quark B-L charge)
coupling_inv = gap**2/K_YY              # gap^2 / Killing = 1/6
print("central_number: charge=", charge, "(=1/3, quark B-L); coupling_inv=", coupling_inv, "
      (=1/6)")

# --- chain validation: reproduce the Higgs quartic ---
lam = 2*sp.sqrt(3)/27
mH = sp.sqrt(2*lam)*sp.Rational(24622,100)
print("quartic_chain: lambda=", float(lam), "m_H=", float(mH), "GeV")

# --- two-branch diagnostic on real colour holonomy ---
Yn = np.array(Y/ sp.sqrt(K_YY), dtype=float)
def diag3(D):
    r=np.random.default_rng(7);P=[r.normal(size=(4,4))+1j*r.normal(size=(4,4)) for _ in range(6)]
    P=[p+p.conj().T for p in P]
    return np.mean([np.linalg.norm(D@p-p@D) for p in P])
def central(D):
    LD=logm(D.astype(complex)); return abs(np.trace(LD.conj().T@Yn)/np.trace(Yn.conj().T@Yn))
Af=lambda i,j:np.array(A(i,j),dtype=float); Sf=lambda i,j:np.array(S(i,j),dtype=float)
th=0.6; Dc=expm(th*Af(0,1))@expm(th*Sf(0,1))@np.linalg.inv(expm(th*Sf(0,1))@expm(th*Af(0,1)))
D_ret=expm(1j*Yn*th)@Dc; D_quo=Dc
print("RETAIN_B-L: [D,H] || =%.2f central#=%.3f"%(diag3(D_ret),central(D_ret)))
print("QUOTIENT_B-L: [D,H] || =%.2f central#=%.3f"%(diag3(D_quo),central(D_quo)))

```

C Companion documents

where_my_hunches_went.md (narrative letter) · operator_constraints.md (constraint list C1–C12) · uniformity_addendum.md (conditional-uniformity full statement) · admissibility_note.md (filter, two failures, coupling). Source framework: TR-2026-FF06a, *Colour from Gravity*.

All numbers regenerated from the drivers in Appendices A–B. Negatives kept first-class. Nothing tuned to a known answer. The charge/coupling split (§13) is the organising result: charge-type predictions are forced exactly; coupling-type predictions carry an EW-scale $\sim 1\%$ normalisation residual.